LESSON PLAN

ROCKET ENGINE (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Advantages and disadvantages of composites
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Advantages and disadvantages of composites. Difference between Isotropic, Anisotropic and Orthotropic materials.
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the development of composite materials, types and functions of fibers and resin, Defects in composite, selection criteria for Matrix, fabric weaves, prepress, strength flowcharts

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of composite strength flowcharts and fabric weave patterns; Instructor supervision required for material property identification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary components of a composite material?
(b) Name one advantage of using composites over traditional metals.
(c) Define the term 'matrix' in the context of composites.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Development of composite materials • Types of fibers and resin functions • Advantages and disadvantages of composites • Defects in composite structures	L & Ds	PPT, IB	15 min
(b)	• Difference between Isotropic materials • Difference between Anisotropic materials • Difference between Orthotropic materials • Selection criteria for Matrix and fabric weaves • Prepress and strength flowcharts	L & Ds	PPT, PDF	20 min
(c)	• Review of composite selection criteria • Analysis of strength flowcharts • Summary of defects and mitigation • Fabric weave patterns overview • Matrix selection guidelines	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the key difference between isotropic and anisotropic materials.
(b) List three major advantages and two disadvantages of composite materials.
(c) How does fabric weave affect the mechanical properties of a composite?

12. Home Assignment and Follow Up	Prepare a brief note comparing the structural applications of isotropic metals versus anisotropic composites in aircraft construction, referencing LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Advanced composite i.e. carbon-carbon composite, metal-matrix composite and their utilization.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min